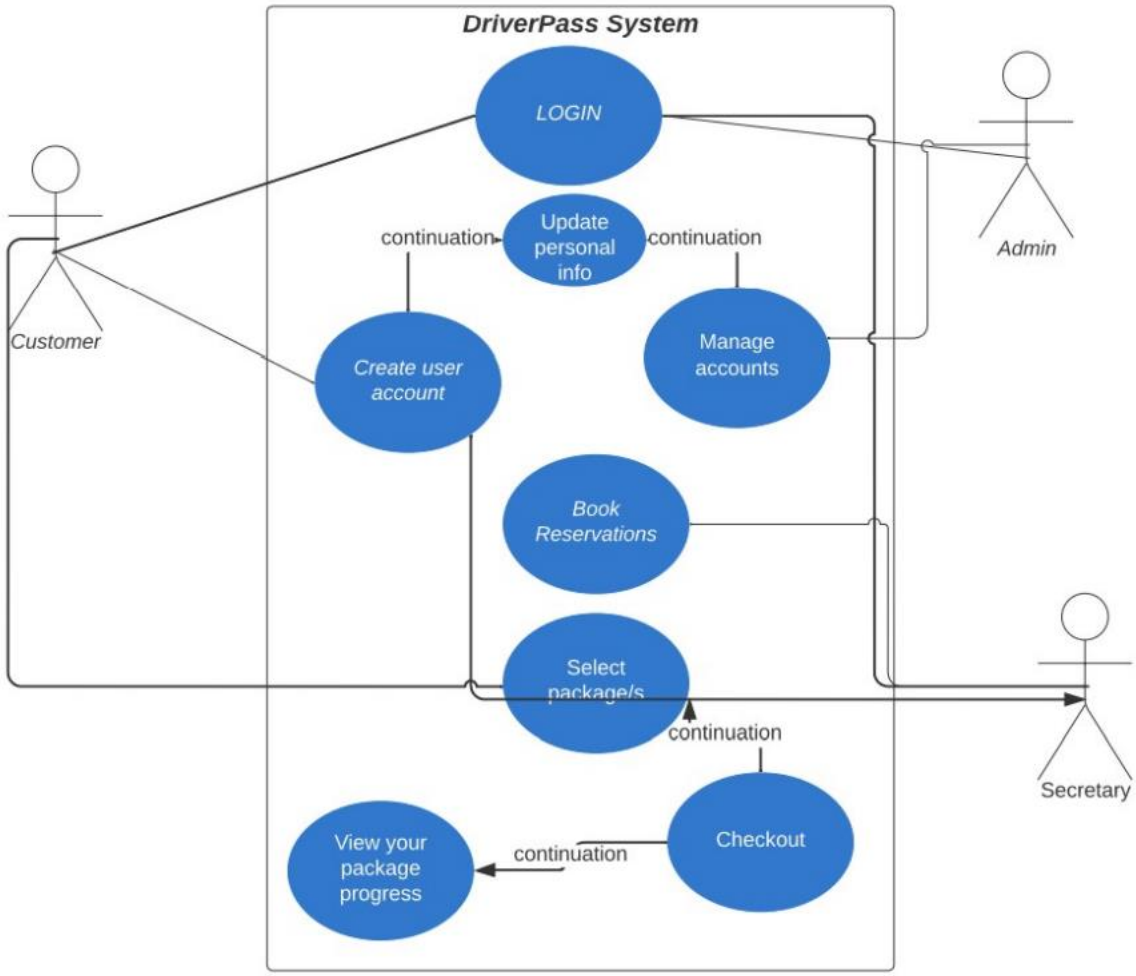
Southern New Hampshire University

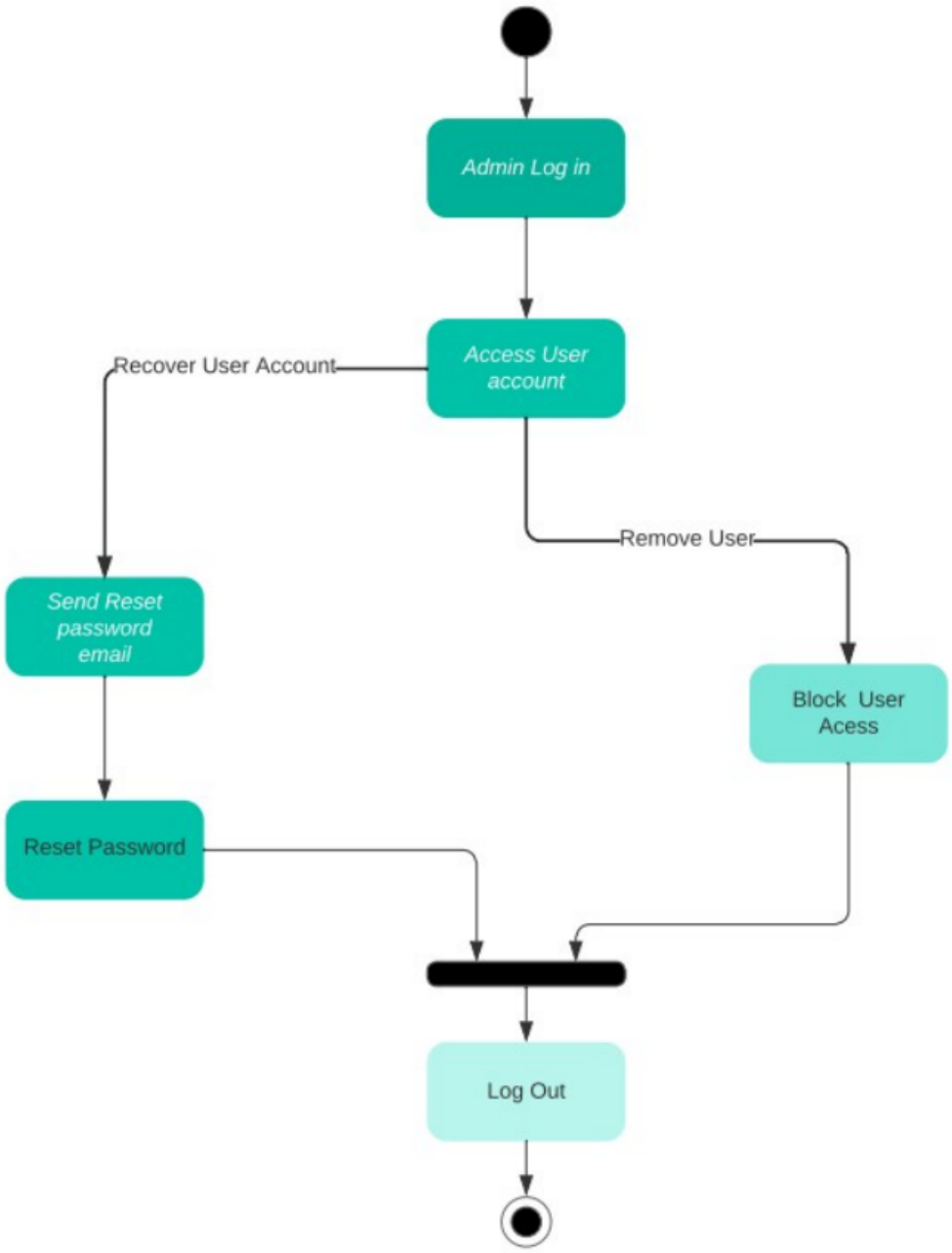
# CS 255 System Design Document

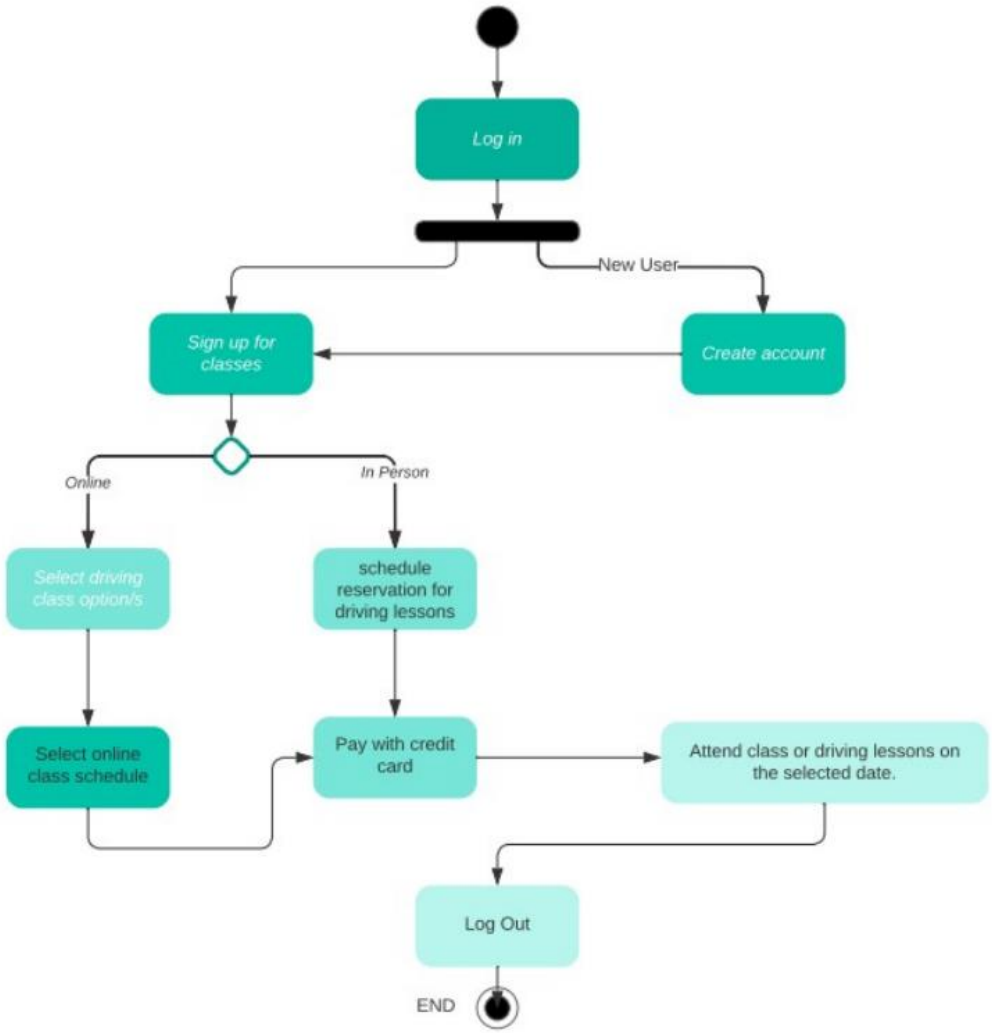
## UML Diagrams

### UML Use Case Diagram

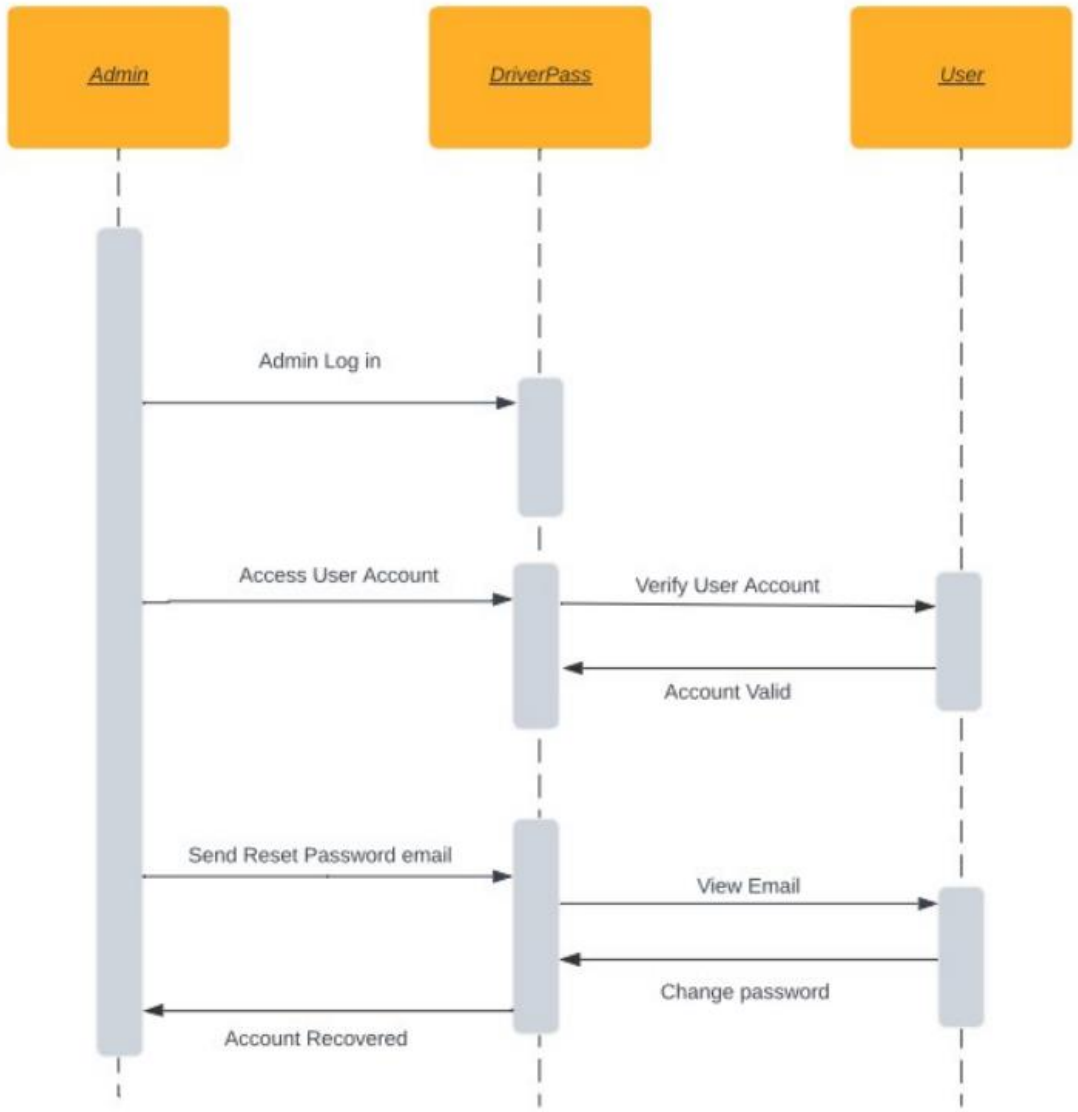


**UML Activity Diagrams**

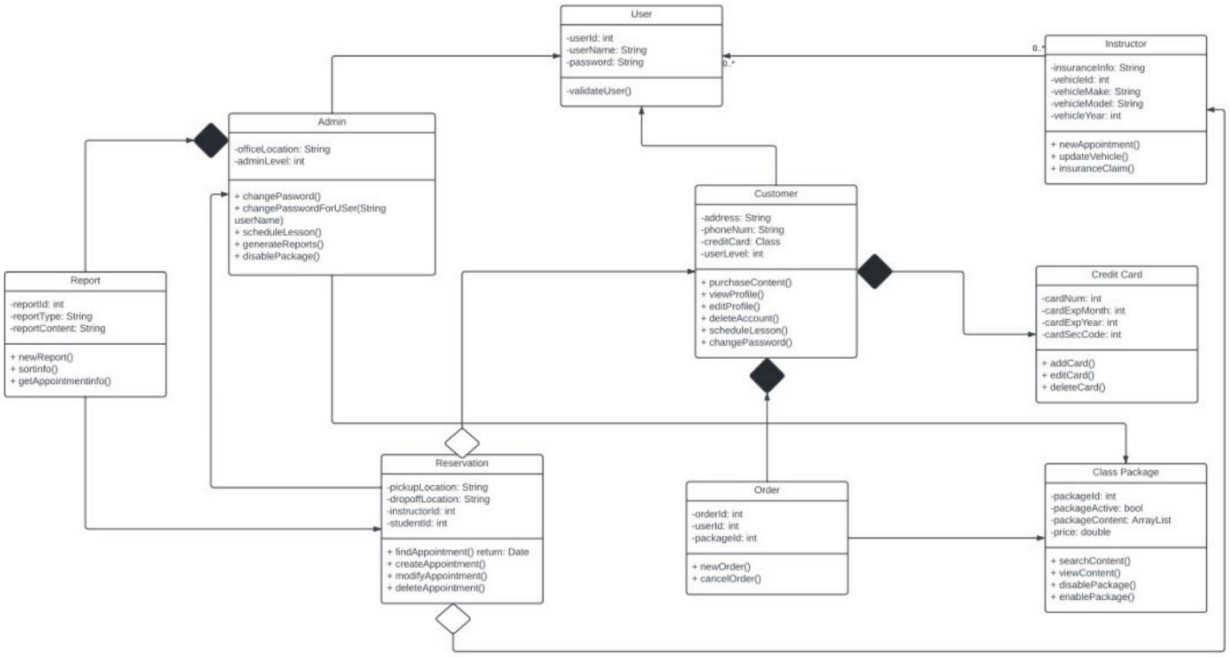



Log in
New User
Sign up for classes
Create account
Online
In Person
Select driving class option/s
schedule reservation for driving lessons
Select online class schedule
Pay with credit card
Attend class or driving lessons on the selected date.
Log Out
END

**UML Sequence Diagram**



**UML Class Diagram**




## Technical Requirements

Based on the system design and diagrams, the following technical requirements are identified:

**Hardware Requirements**

- Cloud-hosted servers for scalability and availability
- Secure database servers for user and transaction data
- Admin workstations for system management

**Software Requirements**

- Web-based application accessible via modern browsers
- Backend server framework (e.g., REST-based architecture)
- Secure relational database (e.g., MySQL or PostgreSQL)
- Integration with third-party payment gateways

**Tools and Infrastructure**

- CASE tools such as Lucidchart for design and documentation
- Secure email services for notifications and password recovery

- Role-based access control system
- Encryption for sensitive data (passwords, payment info)

These requirements ensure that the DriverPass system is secure, scalable, and reliable.